ADVANCED PRACTICE EXAM 4

ANSWER KEY WITH WORKED EXPLANATIONS

Score each single-choice question 1 point. Multi-select ([SELECT TWO/THREE]) and [BUILD LIST] items score 1 point only if completely correct - no partial credit, like the real exam.

QUICK MARKING TABLE

Q1: C	Q2: D	Q3: B	Q4: A	Q5: A	Q6: D
Q7: C	Q8: D	Q9: A	Q10: D	Q11: C	Q12: A
Q13: D	Q14: C	Q15: D	Q16: A	Q17: D	Q18: B
Q19: A	Q20: B	Q21: A, B	Q22: C	Q23: D	Q24: C
Q25: A, C	Q26: D	Q27: B	Q28: A, B	Q29: A	Q30: C
Q31: E-B-D-A-C-F	Q32: D	Q33: A, B	Q34: B, C	Q35: B	Q36: B-D-A-C
Q37: C	Q38: A, B	Q39: A	Q40: D	Q41: C, D	Q42: A
Q43: B	Q44: D	Q45: C	Q46: B	Q47: B, C	Q48: C
Q49: A	Q50: D	Q51: C	Q52: B	Q53: D	Q54: B
Q55: B	Q56: C	Q57: B	Q58: C	Q59: C	Q60: D

EXPLANATIONS - SECTION A - CALCULATION TRACES: DATES, STRINGS, NUMBERS, LOGIC (Questions 1-12)

Question 1 - Correct: C

DATEDIFF counts date-part BOUNDARIES crossed between the two dates, not elapsed duration. With weeks starting on Sunday, the Sundays crossed between 2024-01-15 and 2024-03-03 are: Jan 21, Jan 28, Feb 4, Feb 11, Feb 18, Feb 25, and Mar 3 = 7.

The 48-day gap tempts an elapsed-weeks answer of 6.8 (option A uses that logic); boundary counting is the correct mental model for every DATEDIFF question.

Question 2 - Correct: D

DATEADD preserves the day-of-month when it is valid in the target month: June 30 + 6 months = December 30. Tableau only clamps to the end of the month when the target day does NOT exist (e.g., Jan 31 + 1 month becomes Feb 28/29). Option A misapplies the month-end clamping rule.

Question 3 - Correct: B

MAKEDATETIME(date, time) is purpose-built to combine a date field and a time field into one datetime. DATEPARSE converts strings to dates, '+' will not concatenate date/time parts, and option D builds an unparseable string in most locales.

Question 4 - Correct: A

TRIM removes leading and trailing spaces, leaving "AB", and LEN counts 2 characters. Option B is the untrimmed length; option C assumes only leading spaces were removed; D double-counts the string quotes.

Question 5 - Correct: A

09:00 + 13 hours = 22:00 on the same day. 'hour' (plus 'minute' and 'second') are fully supported date parts in DATEADD, DATEDIFF, DATEPART and DATETRUNC when the field is a datetime.

Question 6 - Correct: D

STR(12345) converts to the string "12345", and LEFT takes the first two characters: "12". There is no error because STR does the conversion first. Note the result is text, not the number 12 (option C), and option A is RIGHT's answer.

Question 7 - Correct: C

FLOOR always rounds DOWN toward negative infinity, so FLOOR(-9.4) = -10. Contrast with INT(-9.4) = -9 (INT truncates toward zero), CEILING(-9.4) = -9, and ROUND(-9.4, 0) = -9. This trio - INT, FLOOR, CEILING on negatives - is a classic exam trap.

Question 8 - Correct: D

AND binds more tightly than OR: the expression reads TRUE OR (FALSE AND FALSE) = TRUE OR FALSE = TRUE, so the row returns "Keep". Always parenthesize mixed logic in real work, but for the exam know that AND wins precedence over OR.

Question 9 - Correct: A

IIF(test, then, else, [unknown]) is a function call - the THEN/ELSE/END syntax belongs to the IF...THEN...END statement form. The editor flags 'The calculation contains errors'. Both forms exist on the exam; swap them deliberately in a couple of practice calcs so the shapes are burned in.

Question 10 - Correct: D

DIV performs INTEGER division: 17 divided by 5 = 3 (the fractional remainder is discarded, not rounded). Option A is ordinary 17 / 5; option C would be a rounding-up error (that is CEILING(3.4)).

Question 11 - Correct: C

CONTAINS is case-SENSITIVE: "TABLEAU" in capitals is not found inside "Tableau Public", so the result is FALSE. A robust pattern is CONTAINS(UPPER([Name]), "TABLEAU") or CONTAINS(LOWER([Name]), "tableau").

Question 12 - Correct: A

YEAR() = 2024 and MONTH() = 10, so 2024 * 100 + 10 = 202410 as a number. Option B concatenates strings and returns TEXT "202410" (wrong data type for numeric sorting/math). Option C adds them into 2034. Option D returns strings ("2024" + "October") because DATENAME always yields text.

EXPLANATIONS - SECTION B - LEVEL OF DETAIL EXPRESSIONS (Questions 13-22)

Question 13 - Correct: D

FIXED LODs are computed BEFORE dimension filters, so the Alpha region total is still 60 + 40 = 100 even though only the Cat B rows (40) are visible. Option A is what a plain SUM([Sales]) shows; option C is the whole-table grand total; the mark is not removed, so NULL is wrong.

Question 14 - Correct: C

The boolean filter calc is itself a FIXED LOD, so it is evaluated at the FIXED stage - before the Category dimension filter. East's full-region total (1,200) beats the threshold; West's (800) does not. The visible Category A figures (300/500) are red herrings. Two lessons: LODs work inside filter calculations, and the filter stage order decides the outcome.

Question 15 - Correct: D

EXCLUDE removes the only dimension in the view, so the LOD is computed at the grand level: the average of ALL underlying rows = (100 + 50 + 200) / 3 = 116.67, broadcast to every mark. Option A is Alpha's own row average; option B uses SUM instead of AVG.

Question 16 - Correct: A

INCLUDE expands the calc to State grain first: NY total = 100 + 100 = 200, NJ total = 300. The outer AVG then averages those STATE totals: (200 + 300) / 2 = 250. Option B is the raw row average (500 / 3), option C is what a SUM wrapper would return, and D forgets NY's second order.

Question 17 - Correct: D

The FIXED LOD broadcasts the customer's LAST order date (Mar 5) onto every row. DATEDIFF('day', Jan 10, Mar 5) counts day boundaries: 21 remaining in January + 29 in February (leap year) + 5 in March = 55. Option B is the March row's value (distance from latest to itself), and option C reverses the argument order. This pattern - 'days since/until the customer's last order' - pairs a LOD with a date function, exactly how the exam composes topics.

Question 18 - Correct: B

Unlike INCLUDE (which starts from the viz grain), FIXED [City] builds city totals independently of the view: Boston = 100 + 60 = 160, Austin = 80. The outer AVG averages those city totals inside the East mark: (160 + 80) / 2 = 120. Option A is the region SUM; option C is the raw row average (240 / 3); D is Boston's value alone.

Question 19 - Correct: A

I needs immunity from view grain and dimension filters - FIXED. II lives at State grain and needs the coarser Region total beside it - EXCLUDE [State] removes the finer State from the computation. III is at Region grain and needs a finer order-level computation averaged up - AVG of INCLUDE [Order ID]. Note II is the trap: INCLUDE [Region] would do nothing useful because Region is coarser than the viz grain.

Question 20 - Correct: B

The inner LOD builds city totals (100, 250, 350, 500); the outer MIN keeps the SMALLEST city total inside the region = 100. Option A answers SUM, option C answers AVG, and D picks an arbitrary middle city. Know the full family: MAX = biggest city (covered in your drill), MIN = smallest, AVG = typical city, SUM = region total.

Question 21 [SELECT TWO] - Correct: A, B

Nested LODs always evaluate inside-out. The inner braces run first (any LOD type may nest inside), and the outer expression aggregates those per-row results. There is no two-level cap in the product.

Question 22 - Correct: C

The inner LOD collapses rows to customer totals (A = 100, B = 300); the outer dimensionless FIXED takes the grand average of those CUSTOMER totals: (100 + 300) / 2 = 200, shown on every mark. Option A is the plain row average (400 / 3) - precisely what this pattern replaces; B is the company total; D is customer A only.

EXPLANATIONS - SECTION C - TABLE CALCULATIONS (Questions 23-30)

Question 23 - Correct: D

The window asks for the previous, current and next mark. April is the last mark, so the window SHRINKS to the available rows (Mar = 70, Apr = 90): (70 + 90) / 2 = 80. WINDOW functions do not return NULL at the edges unless the window is empty. Option A is the average of all four months; C is April alone.

Question 24 - Correct: C

Pane (across) restarts the running total inside each Region pane: East accumulates 10, 30, 70 - so East-Mar = 70 and West-Mar = 35. Option B (110) is what Table (across) would produce by sweeping straight across the pane boundary; A ignores the accumulation; D misreads West.

Question 25 [SELECT TWO] - Correct: A, C

INDEX() numbers the sorted marks 1..12, so INDEX() <= 5 keeps the top five (table-calc filter - remember it hides, not deletes). The Top: By Field option on a dimension filter does the same at the filter stage. Distribution bands shade ranges; bins bucket values - neither ranks members.

Question 26 - Correct: D

'asc' assigns rank 1 to the SMALLEST value - useful for 'worst first' leaderboards. P-B at -20 is the lowest profit, so it ranks 1; 'desc' would rank P-D first and P-B last (4). RANK defaults to 'desc'.

Question 27 - Correct: B

SIZE() returns the number of rows in the current PARTITION. Pane (down) partitions by each Region pane, so East's pane holds 3 rows (SIZE = 3) and West's holds 2. Table (down) would return 5 on every row; FIRST/LAST answer offsets, not counts.

Question 28 [SELECT TWO] - Correct: A, B

Table calculation filters apply AFTER every other filter and after the table itself is computed - they selectively hide marks. That is exactly why a hidden-month % of total still totals 100%. They neither shrink extracts nor restrict the underlying query - that is what measure filters do (option D).

Question 29 - Correct: A

Pane (across) restarts at the pane boundary: West's first month has no earlier marks INSIDE its pane, so the average is just 40. Option B (30) is the Table (across) result, averaging East's 20 and 30 with West's 40 across the boundary - the classic sign your compute-using is wrong.

Question 30 - Correct: C

Percent of Total and TOTAL() are table calculations - they see only the filtered view (800), so East would show 37.5%. The FIXED grand total is computed before dimension filters, keeping the denominator at the full 1,000 and East at 30%. This is the standard 'share of unfiltered total' pattern.

EXPLANATIONS - SECTION D - FILTERS, ORDER OF OPERATIONS, PARAMETERS, SETS & STRUCTURES (Questions 31-36)

Question 31 [BUILD LIST] - Correct: E-B-D-A-C-F

The pipeline: Extract filters (filter at extract creation) - [data source filters] - Context filters - Sets / conditional / Top N (membership is computed before dimension filtering) - Dimension filters - Measure filters - Table calculation filters. The subtle point is that sets and Top N run BEFORE ordinary dimension filters - which is why a raw Top 10 plus a dimension filter surprises people.

Key fact for the exam: when a Top N must run inside a filtered subset, the dimension filter must be promoted to context so it fires earlier than the Top N stage.

Question 32 - Correct: D

Top N is computed BEFORE dimension filters in the order of operations, so the raw Top 10 spans all categories and the Category filter merely hides the eight non-Furniture members. Promoting Category to a context filter moves it before the Top-N stage, giving the Top 10 within Furniture. Option C is fiction - pills on the Filters shelf cannot be re-ordered manually.

Question 33 [SELECT TWO] - Correct: A, B

The Wildcard tab offers Contains/Begins with/Ends with matching; the Condition tab accepts any boolean formula, including CONTAINS. The Top tab ranks members; measure filters test aggregated numbers, not substrings.

Question 34 [SELECT TWO] - Correct: B, C

Hierarchies are purely a Data-pane organization trick: drag-and-drop creation, rename, drill with +/- in the view. They are not limited to geography, and they materialize nothing in the extract.

Question 35 - Correct: B

'Shared members in both sets' is the intersection - the top-100 customers who are also unprofitable. 'All members in both' is the union (everyone in either). 'Except' subtracts. Combined sets work on any two sets over the same dimension.

Question 36 [BUILD LIST] - Correct: B-D-A-C

Create the parameter first, then the CASE calculation that reads it (parameters are global and typed - the calc must reference the exact list strings), then expose the control, then wire the view and verify. The same skeleton serves measure-swaps, dimension-swaps and Top-N parameters - memorize it.

EXPLANATIONS - SECTION E - CONNECT TO & TRANSFORM DATA (Questions 37-48)

Question 37 - Correct: C

A hard 5-minute freshness SLA rules out every refresh-based option (extracts are point-in-time snapshots, and scheduled incremental refresh still lags). The database being fast removes the usual performance counter-argument, and the no-copies policy makes the choice unambiguous: go live.

Question 38 [SELECT TWO] - Correct: A, B

Incremental refresh is an extract-only refresh mode, and the REGEXP_* functions run in Hyper (extracts) - live databases typically offer only vendor-specific equivalents. Custom SQL and Initial SQL are features of LIVE connections; switching a connection to an extract removes them.

Question 39 - Correct: A

Unions stack rows of identically structured tables. Joins need key columns and merge sideways; blends aggregate across sources at the viz level; relationships keep tables separate at the logical layer.

Question 40 - Correct: D

This is the textbook relationship-vs-join case. An inner join stamps the month's target onto every order line, so SUM(Target) is multiplied by the number of lines. A relationship (logical layer) keeps Targets at its own grain: SUM(Target) computes correctly per region-month, and SUM(Sales) per the orders grain is uninflated.

Question 41 [SELECT TWO] - Correct: C, D

Clean steps carry the everyday edits: rename, trim, split, change data type, edit values, filter, remove duplicates. Scheduling belongs to Prep Conductor on Server/Cloud, and joins/unions are their own step types in the flow.

Question 42 - Correct: A

A columns-to-rows pivot always generates 'Pivot Field Names' (the former column headers) and 'Pivot Field Values' (the cell contents) - rename them immediately to something like Quarter and Budget before building views.

Question 43 - Correct: B

Automatic Split would split on EVERY comma and produce three fields (over-splitting 'Jr.'). Custom Split lets you choose the separator AND how many columns to split off - 'first 1' gives Smith and 'John, Jr.'. This is precisely when the exam expects Custom over Automatic.

Question 44 - Correct: D

Aggregate steps reshape grain: drag Region and the month field to Grouped Fields, and the measures (SUM(Sales), COUNTD(Order ID)) to Aggregated Fields. Pivots reshape columns/rows but do not aggregate; Clean steps edit values; joins combine tables.

Question 45 - Correct: C

Prep's Union step detects field mismatches and offers Merge Fields to fold them into one column. (In Desktop there is no merge button - aligning names beforehand is the fix; in Prep it is Merge Fields.)

Question 46 - Correct: B

CSV is the universal hand-off format - any spreadsheet or ETL tool reads it. .hyper/.tdsx/published data sources all require Tableau to open or connect. Output-type choice questions always resolve to who consumes the result and with what tooling.

Question 47 [SELECT TWO] - Correct: B, C

Field default properties include number format, aggregation, color, sort order, comment and fiscal year start - they apply everywhere the field is dropped unless overridden. Schedules live on Server/Cloud, and restricting user-entered values is the job of parameters, not field properties.

Question 48 - Correct: C

Identifiers that are numbers are still categorical: converting to a dimension makes Tableau treat each ID as a distinct label/grouping member instead of an aggregatable measure. Bins bucket ranges and AVERAGE still computes arithmetic on IDs.

EXPLANATIONS - SECTION F - CREATE CONTENT: CHARTS, DASHBOARDS, FORMATTING (Questions 49-56)

Question 49 - Correct: A

A histogram bins a single continuous measure and shows the frequency per bucket - distribution shape made visible. A box plot summarizes quartiles (not per-band counts), Pareto ranks categories, and scatter needs two measures.

Question 50 - Correct: D

Highlight table recipe: Square marks fill each cell, the measure on Color drives the shading, and the same measure stays on Text so the numbers remain visible. Row banding is cosmetic zebra striping, not value-driven color.

Question 51 - Correct: C

Synchronize Axis forces both measures onto one shared scale so the visual proportions are honest - Tableau never synchronizes automatically. Independent axes cause exactly the reported distortion.

Question 52 - Correct: B

Device Preview + an added phone layout lets you compose a custom arrangement per device class; Tableau picks the right layout at render time. Size Range only constrains overall dimensions - it never re-arranges tiles.

Question 53 - Correct: D

Parameter actions write a single clicked value into a parameter, which titles (Insert > Parameter) and calculated fields can reference anywhere in the workbook. Filter actions filter marks; highlight actions emphasize marks; set actions store multi-member membership - none can feed a title string.

Question 54 - Correct: B

The Navigation dashboard object jumps viewers between sheets/dashboards of the same workbook, honoring permissions, all inside the Tableau session. URL actions are built for external links and would round-trip through the browser address bar.

Question 55 - Correct: B

Custom SHAPES live in My Tableau Repository > Shapes (one subfolder per palette); reload via the Edit Shape dialog. Preferences.tps is where custom COLOR palettes are defined - the exam loves pairing these two repository locations as distractors.

Question 56 - Correct: C

A Distribution Band set to Percentiles shades BETWEEN the two percentile values - exactly the inter-quartile middle 50%. Standard deviation bands measure spread around the mean, not quartiles; reference lines draw marks, not shaded regions; confidence intervals belong to trend models.

EXPLANATIONS - SECTION G - PUBLISH & MANAGE ON SERVER / CLOUD (Questions 57-60)

Question 57 - Correct: B

A published data source centralizes connection, metadata and calculated definitions; certification flags it as steward-endorsed. Packaged copies drift apart the day they are emailed - defeating the point.

Question 58 - Correct: C

Subscriptions email an image/PDF snapshot of a view on a schedule. Alerts fire only when a metric crosses a threshold (not calendar-guaranteed), custom views just remember filter state, and extract refresh updates data - it sends nothing.

Question 59 - Correct: C

Only Creator can CONNECT TO DATA and perform full web authoring (and publish flows/raw sources). Explorer (can publish) can edit and republish existing content built on existing published data sources but cannot create new connections; Viewer is consume-only; the admin Explorer variant still caps at Explorer capabilities.

Question 60 - Correct: D

Custom views store a per-user snapshot of filter/selection state on published content and can be saved privately or shared. Subscriptions email snapshots, duplication forks content, and quality warnings flag stale/broken data.

SECTION WEIGHTING REMINDER

This paper deliberately over-weights your weak areas: Sections A-D (calculation traces, LODs, table calcs, filters/parameters) = 36 questions versus about 25 on the real exam. If you scored below 70% inside Sections A-D, redo the calculations drill and advanced papers 1-3 keys before your next attempt. Sections E-G carry the remaining 24 questions (connections/Prep, content creation, publishing) mapped to the 24/26/9 weightings.